

Trader Behavior Analysis Based on Bitcoin Market Sentiment

Introduction

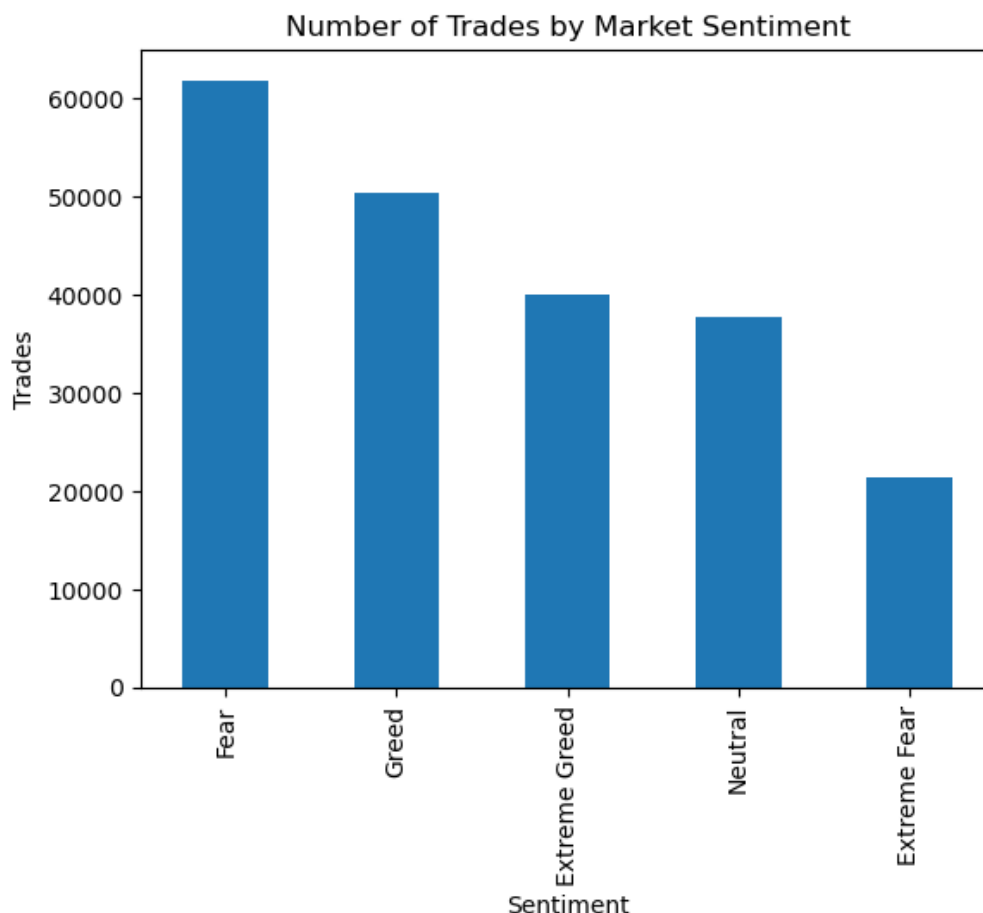
In this project, I analyzed the relationship between Bitcoin market sentiment and trader performance. For this analysis, two datasets were used. The first dataset contains Bitcoin Fear and Greed Index values, and the second dataset contains historical trading data from Hyperliquid.

The main objective of this project was to understand how market sentiment affects trading behavior and profitability. Different market conditions such as Fear, Greed, Extreme Fear, Extreme Greed, and Neutral were compared with trading performance.

Data Preparation

First, both datasets were loaded into Python using Pandas. The date columns were converted into a proper datetime format so that both datasets could be merged correctly.

After cleaning and formatting the data, the datasets were merged using the date column. This allowed each trade to be associated with the corresponding market sentiment for that day.



Analysis and Findings

1. Trading Activity

The highest number of trades were recorded during the Fear market condition. More than 61,000 trades were executed during this period.

This shows that traders remained active even when the market sentiment was negative. On the other hand, the lowest number of trades were observed during Extreme Fear conditions.

2. Average Profit Analysis

The average profit per trade was highest during Extreme Greed conditions.

Extreme Greed: 67.89

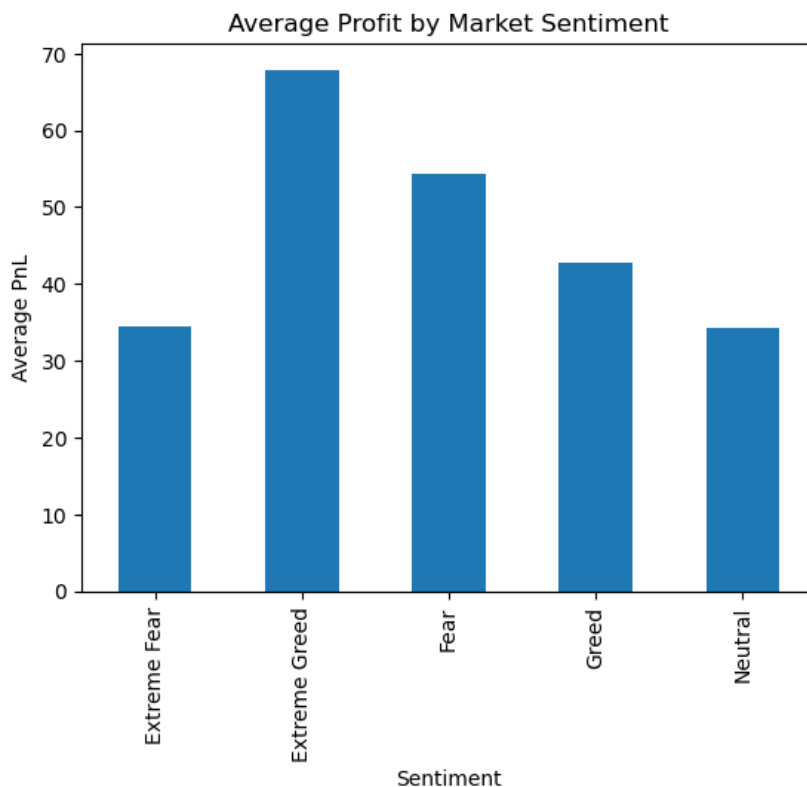
Fear: 54.29

Greed: 42.74

Extreme Fear: 34.54

Neutral: 34.31

This indicates that traders generally achieved better results when market sentiment was strongly positive.



3. Total Profit Analysis

Fear conditions generated the highest total profit among all sentiment categories.

This may be because a large number of trades were executed during Fear periods, which increased the total profit generated by traders.

4. Win Rate Analysis

The highest win rate was observed during Extreme Greed conditions.

Sentiment	Win Rate
Extreme Greed	46.49%
Fear	42.08%
Neutral	39.70%
Greed	38.48%
Extreme Fear	37.06%

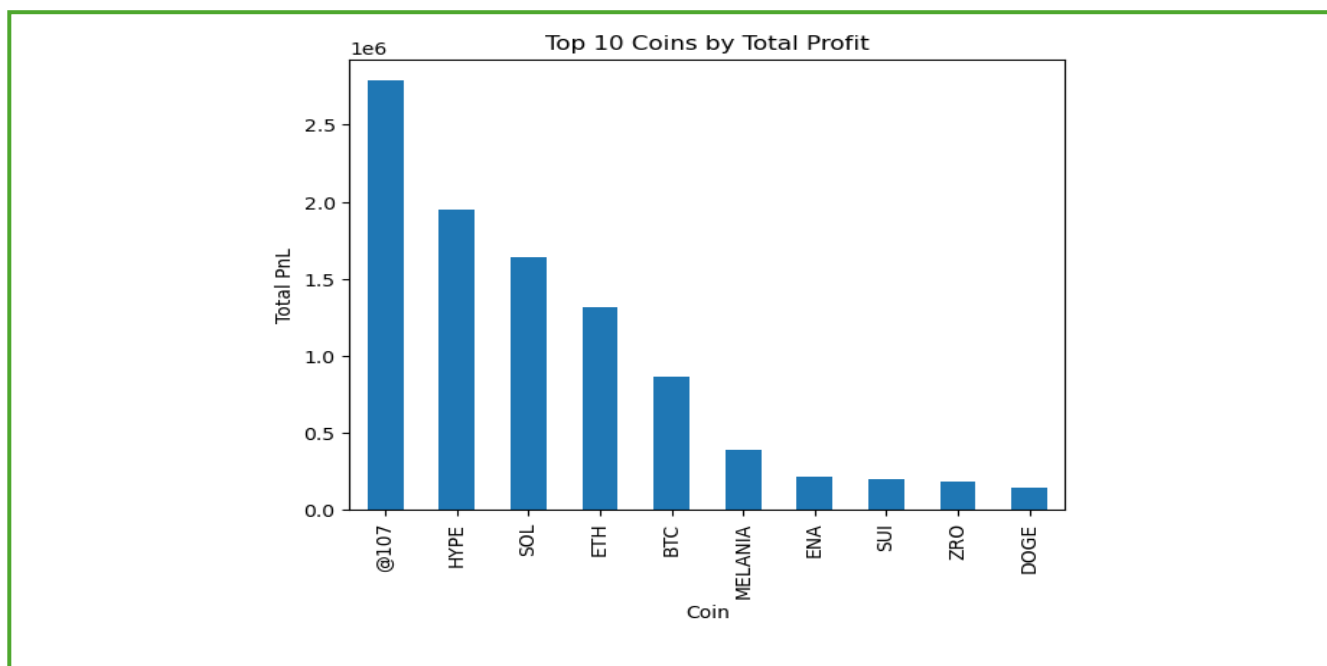
These results suggest that traders had a higher probability of making profitable trades when market sentiment was highly optimistic.

5. Top Performing Coins

The most profitable assets in the dataset were:

- @107
- HYPE
- SOL
- ETH
- BTC

Among these assets, HYPE and SOL generated significant profits and performed better than many other cryptocurrencies.



6. Top Performing Traders

A small number of traders generated a large portion of the overall profit. The highest-performing account earned more than 2 million in total profit.

This suggests that experienced traders may have used better risk management and trading strategies.

Conclusion

From this analysis, it can be concluded that market sentiment has a noticeable impact on trader performance.

Extreme Greed conditions showed the highest average profit and win rate, while Extreme Fear conditions showed weaker performance. Fear periods generated the highest total profit because trading activity was higher during those periods.

Overall, the results indicate that market sentiment can be a useful factor for understanding trader behavior and improving trading decisions.